

ALOK RAJPUT

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EDUCATION

- National Institute of Technology, Silchar** Assam, India
Bachelor of Technology - Electrical Engineering; CGPA: 9.28 Nov 2022 - June 2026
- BNSD Shiksha Niketan Inter College, Kanpur** Uttar Pradesh, India
Intermediate-UP Board; Percentage: 83.8 April 2019 - May 2021
- BNSD Shiksha Niketan Inter College, Kanpur** Uttar Pradesh, India
High School-UP Board; Percentage: 92.8 April 2018 - May 2019

INTERNSHIPS

- 1 × 660 MW Panki Extension, Panki Kanpur** Uttar Pradesh, India
Power Generation and Switchyard Analysis Intern May 2024 - July 2024
 - Studied 660 MW thermal power plant operation with emphasis on steam turbine-based electricity generation.
 - Analyzed switchyard components including busbars, circuit breakers, and isolators for high-voltage transmission.
 - Modeled transmission network and performed load flow analysis in ETAP Grid to assess system stability.
- Technical Project Intern, IIT Guwahati** Assam, India
Speech-Based Mental State Detection using BiGRU Model May 2025 - July 2025
 - Developed **BiGRU-based deep learning model** for speech classification into stressed, depressed, or normal categories.
 - Implemented MFCCs, silence trimming, and noise augmentation techniques.
 - Achieved **88% validation accuracy** using Librosa and Python.

PROJECTS

- Two-Area Power System Load Frequency Control**
MATLAB Simulation Project
 - Modeled and simulated a **two-area interconnected power system** in **MATLAB** to analyze **steady-state frequency response** under step load disturbances using **Load Frequency Control (LFC)**.
 - Studied the impact of **governor, turbine, generator, damping, and tie-line dynamics** on frequency regulation and system stability.
- Virtual Polarizer Dehazing for Underwater Images**
Final Year Project
 - Developed a **Physics-based Virtual Polarizer model** to reduce haze and color distortion in underwater images.
 - Integrated additional modules such as **white balancing, image sharpening** for enhanced visibility.
 - Implemented on **Google Colab** using Python (**OpenCV, NumPy, Matplotlib**); benchmarked performance with **UCIQM, UIQM, and Image Complexity**.

SKILLS

- Languages:** SQL, C, C++
- Platform and Tools:** MATLAB, Simulink, PSpice, MI Power, Google colob, Power BI, Excel
- Domain Knowledge:** Power System, Electrical Machine, Control System, Signal Processing
- Soft Skills:** Event Management, Problem Solving, Data Interpretation, Public Speaking, Cross-functional Communication

POSITION OF RESPONSIBILITY

- AryaBhatta Hostel, NIT Silchar**
Assistant Prefect Nov 2022 - July 2023
 - Assisted hostel administration and addressed student concerns to the concerned authorities.
 - Organized **Saraswati Puja** and led a student team for the **Republic Day parade**, ensuring smooth planning and execution.
- Institution of Engineers (India) Student Chapter, NIT Silchar**
Executive Member Aug 2023 - May 2025
 - Managed and organized multiple workshops and onboarded a new module in **Tecnoesis** (first-time inclusion).
- Incandescence 2025, NIT Silchar**
Co-Head, Management Team
 - Co-led the management team to plan, coordinate, and execute large-scale cultural fest operations.
 - Oversaw logistics, team coordination, and smooth execution of events involving large student participation.

ACHIEVEMENTS/OTHER INTEREST

- Secured 1st position in Logic Wizard and 2nd position in RC car racing in Tecnoesis 2024.
- Selected for the Integrated Scholarship of Uttar Pradesh, among the top 15 students statewide(2018).
- Hobbies include wall painting, Pen Sketching and play ing badminton, Learning Data Analytics, and Spanish as an additional language.